Week 4 Experiment Report

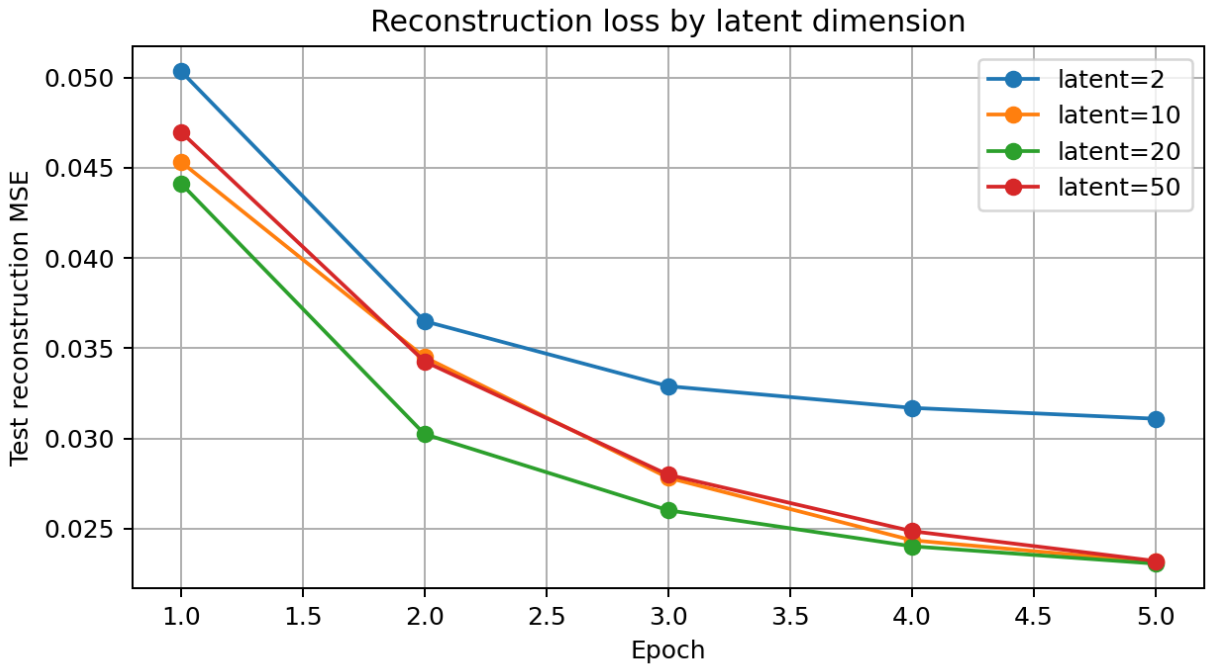
Autoencoder representations and clustering comparison

What was delivered

This week contains two notebooks: one for training an autoencoder and extracting latent features, and one for comparing clustering in raw-pixel, PCA, and AE latent spaces. The comparison table and figures are saved in the week4/results folder.

Experiment setup

The autoencoder used the structure 784 → 256 → 64 → latent → 64 → 256 → 784, MSE reconstruction loss, Adam with learning rate 0.001, batch size 256, and five epochs. To keep the experiment reproducible and practical on a laptop, I used 20,000 training images and 10,000 test images selected with random seed 42.



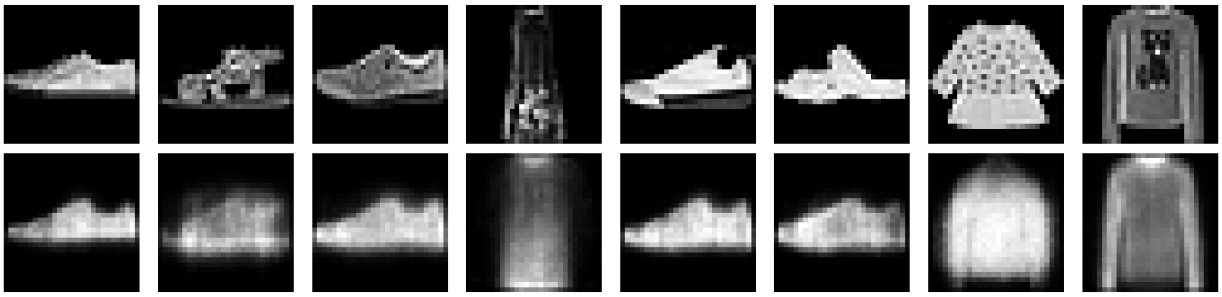
The reconstruction loss generally decreased as the latent dimension increased. However, reconstruction quality and clustering quality did not always improve together.

Representation comparison

The following table compares K-means with K=10. NMI and ARI were calculated after clustering, using the true labels only for evaluation.

Representation	Dimension	NMI	ARI
Raw pixels	784	0.5145	0.3497
PCA-20	20	0.5138	0.3506

In this run, raw pixels and PCA-20 produced higher clustering scores than AE-20. This is a useful result because it shows that a representation optimized for reconstruction is not automatically the best representation for clustering.

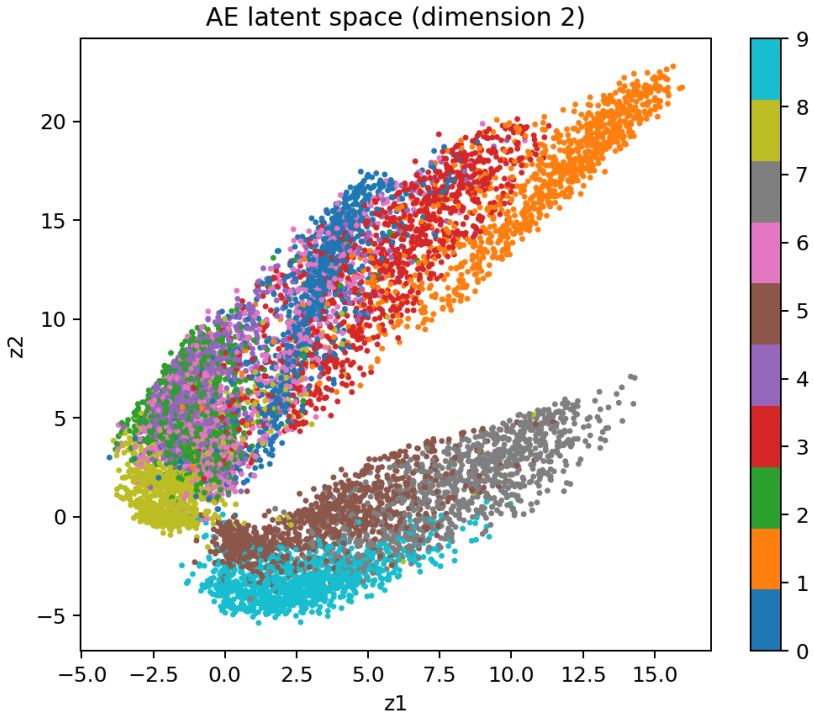


Latent dimension study

The table below records the reconstruction loss and clustering metrics for four autoencoders.

Latent dimension	Reconstruction loss	NMI	ARI
2	0.0311	0.5007	0.3278
10	0.0232	0.4568	0.2654
20	0.0231	0.4471	0.2633
50	0.0232	0.4712	0.2811

The two-dimensional model had the highest AE NMI and ARI among the four AE settings in this short run, even though its reconstruction loss was worse. The 10- and 20-dimensional models reconstructed better but produced lower clustering scores. The 50-dimensional model recovered some clustering performance.



Week 4 questions and analysis

1. What is representation learning? It is learning a new feature representation from data instead of using the original inputs directly.

2. What is a latent representation? It is the lower-dimensional vector produced by the encoder.

3. What is an encoder? The encoder maps an input image to its latent vector.

4. What is a decoder? The decoder maps the latent vector back to a reconstructed image.

5. What is reconstruction? Reconstruction is the decoder output that tries to reproduce the original input.

6. What is a bottleneck? The bottleneck is the restricted latent layer that forces the model to keep the most useful information.

7. Why might AE features work better than raw pixels? An AE can remove some redundancy and encode higher-level structure, although this depends on the training objective.

8. Can good reconstruction still give poor clustering? Yes. Reconstruction preserves information useful for rebuilding pixels, but that information may not separate semantic classes. The measured AE-20 result was below raw pixels and PCA-20.

9. What happens when the latent dimension is very small? Reconstruction becomes harder because the bottleneck is tight, but the latent space may become compact and sometimes easier to visualize or cluster.

10. What happens when it becomes very large? Reconstruction usually becomes easier, but the representation may keep redundant or nuisance details and clustering may not improve.

Main conclusion: better reconstruction does not always mean better clustering. Both objectives need to be measured separately.